

P8139
Homework #5
Due 04/28/26

1. Describe your progress in getting the data for your final project. Please summarize the following information: (no need to have everything ready by the due date, but partial), cancer types, sample sizes, omics data types, study designs, and research questions if your data will come from TCGA. If your data will come from other places, please describe accordingly.

Guideline for the final project (from the Syllabus)

- Each student will do a final project of his/her choice. Final reports are due on Friday 5/15/25 by 5pm. The format of the final report is single-spaced, font size 11 and no more than 10 pages, with figures, tables and references included, like a journal article.
- Data sources:
You can use TCGAbiolinks or other tools to download omics data for your final projects from the TCGA (the Cancer Genome Atlas) project using the 'GDCquery' and 'GDCdownload' functions. (https://bioconductor.org/packages/release/bioc/vignettes/TCGAbiolinks/inst/doc/download_prepare.html)
- For the data you are going to use for your final project, you need to pay attention to the following items:
 - (a) What is the experimental design?
 - (b) What are the samples used? Tumor tissues and adjacent-normal tissues, i.e., paired case-control samples? Or tumor tissues and independent normal tissues, i.e., independent case-control samples? What are the sample size?
 - (c) What are the necessary data preprocessing steps for DNA methylation data? Google to figure this out yourself. If so, list the detailed process steps.
 - (d) What is your research question? What is your data analysis plan? That is, what tests are you going to use to answer your research questions? Two-sample t-test? Paired t-test? Regressions? Are there multiple comparisons issues that you need to take care of? How do you plan to handle that?
 - (e) How to make figures and tables out of these high-dimensional data to best present your results?
 - (f) What is your conclusion?